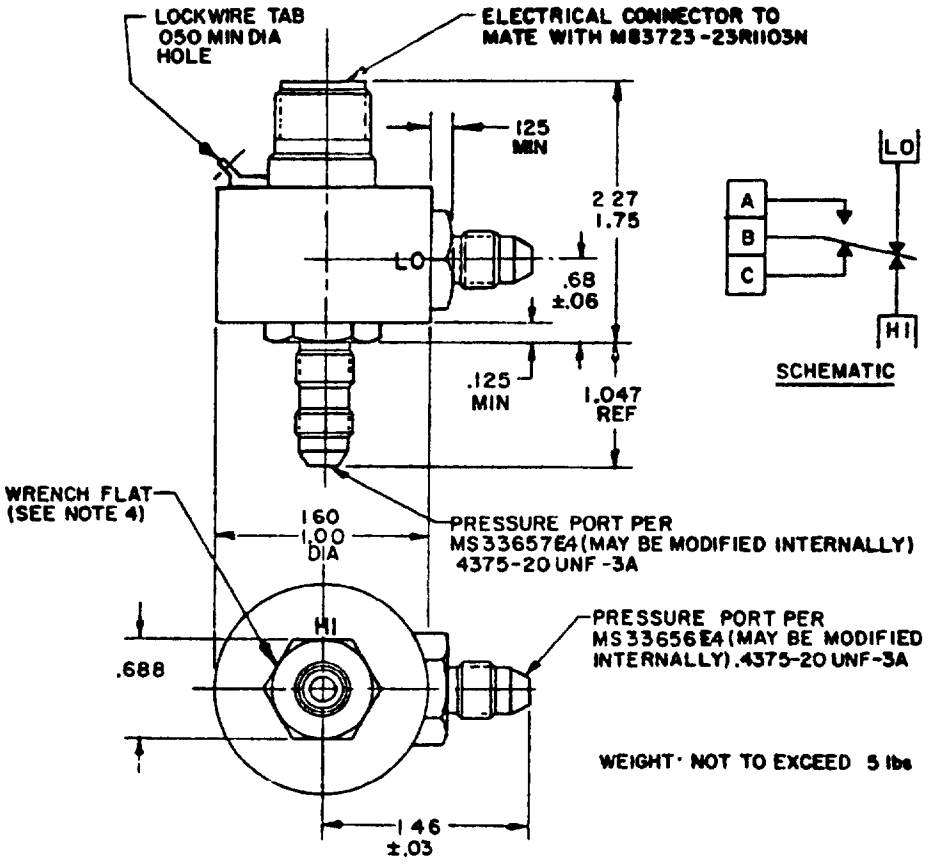
MIL-S-9395/29A
6 OCT 1975
SUPERSEDING
MIL-S-9395/29(USAF)
18 September 1974

MILITARY SPECIFICATION SHEET

SWITCHES, PRESSURE, (DIFFERENTIAL) (TYPE III), 4 AMPERES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the pressure switches described herein shall consist of this document and the latest issue of Specification MIL-S-9395

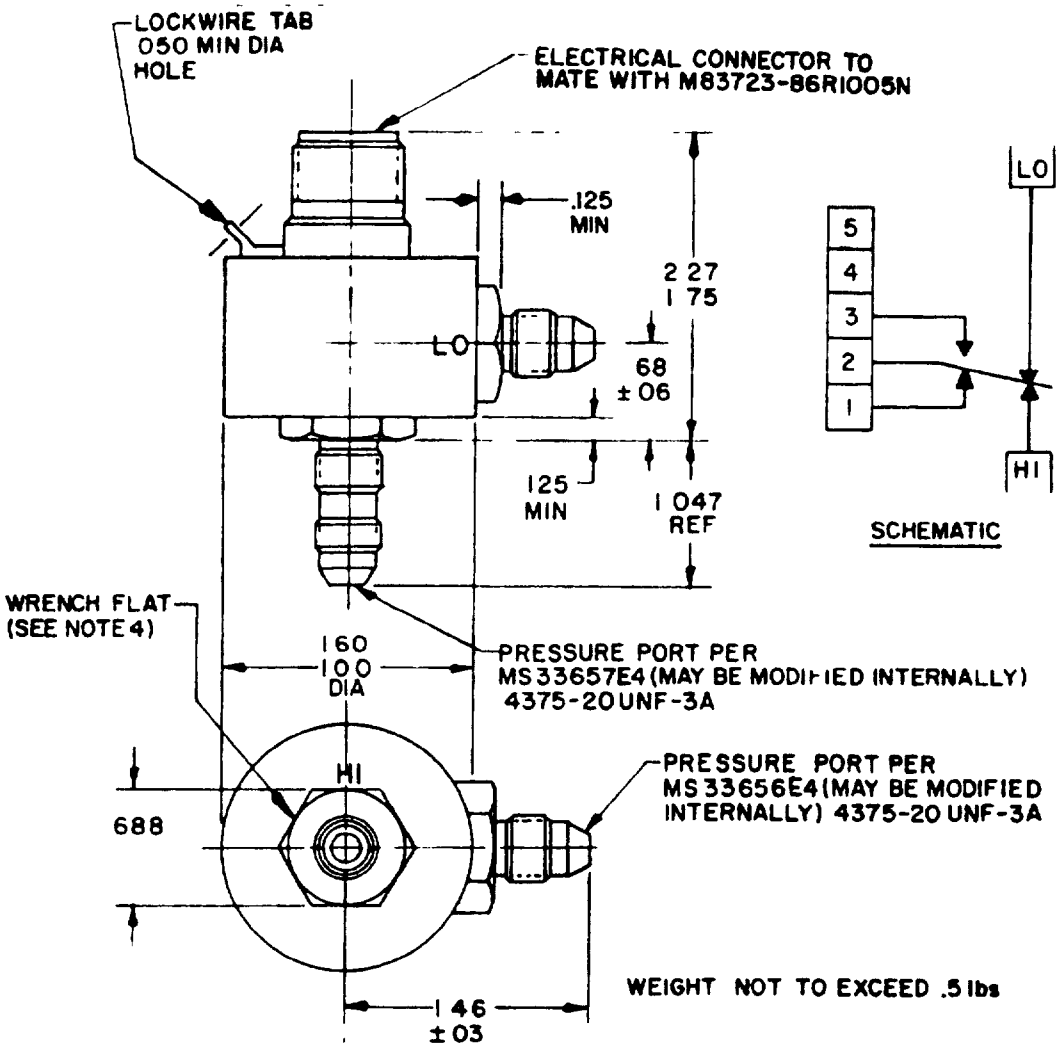


Ⓐ Configuration 1

Ⓐ denotes changes

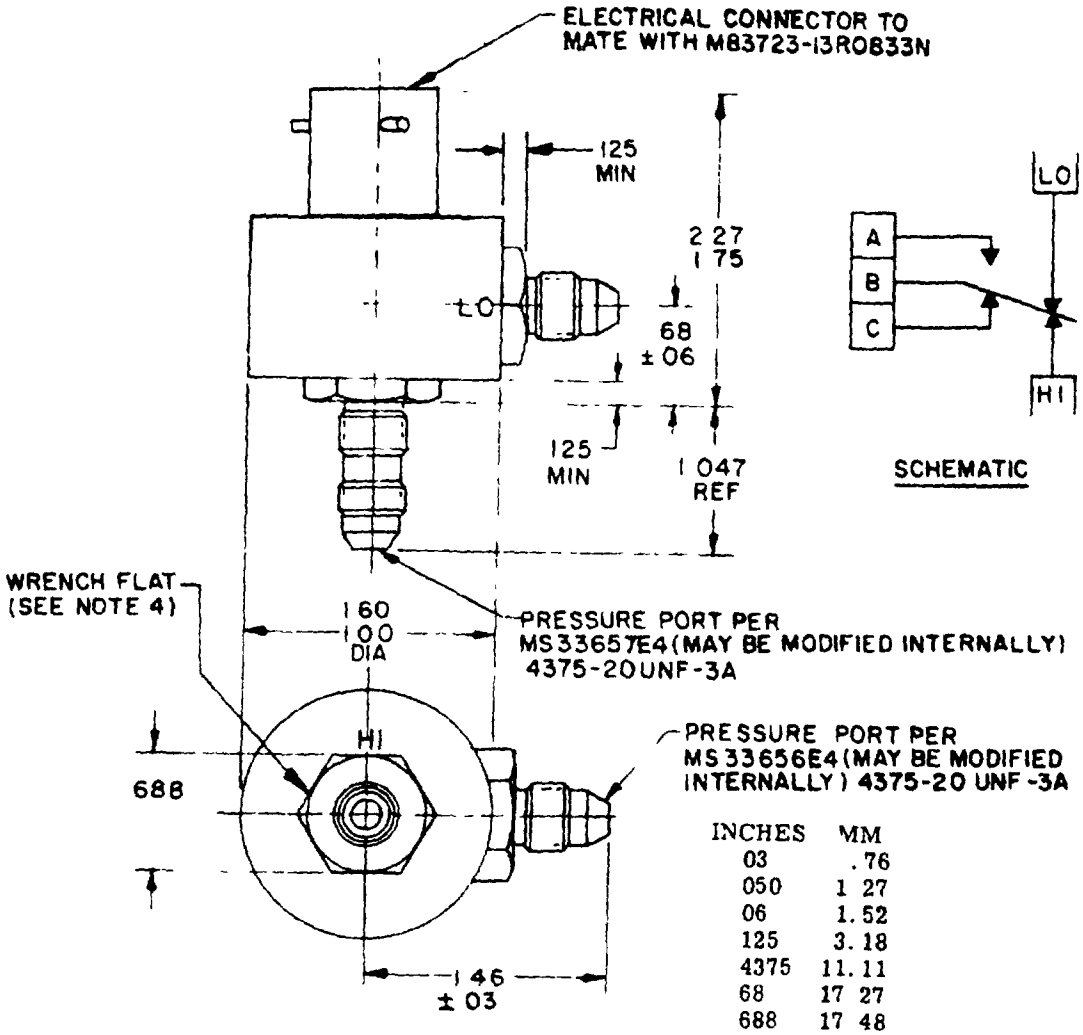
FSC 5930

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Ⓐ Configuration 2

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INCHES	MM
03	.76
050	1 27
06	1.52
125	3.18
4375	11.11
68	17 27
688	17 48
1 00	25.40
1.047	26 59
1 46	37.08
1 60	40.64
1 75	44.45
2 27	56 57

WEIGHT NOT TO EXCEED 5 lbs

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. A minimum of two wrench flats is needed.
5. Ports permanently marked as shown.

Ⓐ Configuration 3

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REQUIREMENTS

Ⓐ Dimensions, weight and electrical schematic See configurations

Calibration See tables I, II, III, and IV

Proof pressure 250 psig high over low and simultaneously

System pressure 125 psig high over low and simultaneously

Burst pressure 500 psig high over low and simultaneously

Electrical ratings
Operating voltage - 28 Vdc
Current rating - 4 amperes resistive
2 amperes inductive

Ⓐ Seal
High pressure chamber Media proof During this test the low pressure chamber shall be open to atmosphere Subject switches to proof pressure for 2 minutes using hydraulic fluid per MIL-H-5606 with chamber pressure continuously being monitored Isolate the chamber at proof pressure with the chamber disconnected from the pressure source Under that condition, the pressure shall not drop more than 1 psig for the first 30 seconds to allow stabilization of test equipment, no pressure loss is allowed thereafter
Low pressure chamber Media proof Test as above except low pressure port is connected to high pressure port
Electrical chamber See table I

Ⓐ Electrical connector See configurations

Ⓐ Pressure port See configurations

Media Dry air, nitrogen, fuel per MIL-T-5624, oil per MIL-L-7808, or hydraulic fluid per MIL-H-5606

High temperature (operating and nonoperating) D (400°F and 500°F for 30 minutes)

Low temperature (operating and nonoperating) D (-65°F)

Altitude D (100,000 feet)

Shock C (100 G)

Vibration S (test condition D, method 204 of MIL-STD-202)

Life mechanical A (100,000 cycles)

Life electrical C (50,000 cycles)

Acceleration C (8 G)

Pulsation amplitude A (0%).

Pulsation frequency A (0 Hz)

Pressure surge B (more than 1,000 psi/sec)

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Dielectric withstanding voltage (at reduced barometric pressure) Applicable

Connector torque. Applicable.

Flame test Applicable

Explosion Applicable

QUALIFICATION

Single submission Restricted to switch submitted
Group submission See table V

PART NUMBER Consists of "M" prefix followed by specification sheet number, a dash (-), and a five-letter code The five-letter code identifies the configuration and pressure setting mode (code from table I), high-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III), and low-pressure setting (coarse value code from table II) followed by fine value with applicable tolerance (code from table III) The five-letter code used in the following example identifies a switch of configuration 1 which actuates on increasing pressure at 1½ ±½ psid, and deactuates on decreasing pressure at ½ ± psid

EXAMPLE

M9395/29- A A D A B

"M" PREFIX AND SPECIFICATION SHEET NUMBER

CONFIGURATION AND PRESSURE SETTING MODE = Code "A" of table I

HIGH-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "D" of table III (fine value of 1½ ±½ psid)

LOW-PRESSURE SETTING = Code "A" of table II (coarse value of 0 psid) and Code "B" of table III (fine value of ½ ±½ psid)

NOTE Design limitations (actuation values and tolerances, deadband and deactuation values and tolerances) should be coordinated with manufacturer(s) listed on the QPL for this specification sheet before specifying a particular "M" number The fact that operating characteristics can be coded does not necessarily mean that it can be manufactured or procured

TABLE I Codes for combinations of configurations and pressure setting modes

	Configuration 1		Configuration 2		Configuration 3		Pressure setting mode	
	Electrical chamber		Electrical chamber		Electrical chamber		High setting	Low setting
	Hermetic	Unsealed	Hermetic	Unsealed	Hermetic	Unsealed		
Code - - -	A	D	G	K	N	R	At (or max) 1/	At (or min) 1/
Code - - -	B	E	H	L	P	S	At (or max) 1/	Differential 2/
Code - - -	C	F	J	M	Q	T	Differential 2/	At (or min) 1/

1/ Setting values are designated by codes from tables II and III
2/ Setting values are designated by codes from table IV

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TABLE II Codes for coarse settings

Code	Coarse value (in psid)	Code	Coarse value (in psid)	Code	Coarse value (in psid)
A - - -	0	L - - -	30	W - - -	80
B - - -	2½	M - - -	35	X - - -	85
C - - -	5	N - - -	40	Y - - -	90
D - - -	7½	P - - -	45	Z - - -	95
E - - -	10	Q - - -	50	1 - - -	100
F - - -	12½	R - - -	55	2 - - -	105
G - - -	15	S - - -	60	3 - - -	110
H - - -	17½	T - - -	65	4 - - -	115
J - - -	20	U - - -	70	5 - - -	120
K - - -	25	V - - -	75	6 - - -	125

TABLE III Codes for combinations of fine settings and tolerance values

	Fine value (in psid) for settings below 20 psid					Tolerance
	0	½	1	1½	2	
Code - - -	A	B	C	D	E	±½ psid
Code - - -	F	G	H	J	K	±1 psid
Code - - -	L	M	N	P	Q	±2 psid
Code - - -	R	S	T	U	V	±3 psid
Code - - -	W	X	Y	Z	1	±5 psid
Code - - -	2	3	4	5	6	±8 psid
Code - - -	7	8	9	0	+	Min or Max
	Fine value (in psid) for settings of 20 psid and above					Tolerance
	0	1	2	3	4	
Code - - -	A	B	C	D	E	±1 psid 1/
Code - - -	F	G	H	J	K	±2 psid 2/
Code - - -	L	M	N	P	Q	±3 psid 3/
Code - - -	R	S	T	U	V	±5 psid
Code - - -	W	X	Y	Z	1	±8 psid
Code - - -	2	3	4	5	6	±15 psid
Code - - -	7	8	9	0	+	Min or Max

1/ Not applicable for pressure settings above 33 psid
2/ Not applicable for pressure settings above 66 psid.
3/ Not applicable for pressure settings above 100 psid

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TABLE IV Codes for differential settings

Code	Differential value (in psid)	Code	Differential value (in psid)
A - - -	0	T - - -	11
B - - -	½	U - - -	12
C - - -	1	V - - -	13
D - - -	1½	W - - -	14
E - - -	2	X - - -	15
F - - -	2½	Y - - -	16
G - - -	3	Z - - -	18
H - - -	3½	1 - - -	20
J - - -	4	2 - - -	22
K - - -	4½	3 - - -	24
L - - -	5	4 - - -	26
M - - -	5½	5 - - -	28
N - - -	6	6 - - -	30
P - - -	7	7 - - -	35
Q - - -	8	8 - - -	40
R - - -	9	9 - - -	45
S - - -	10	0 - - -	50

TABLE V Extent of qualification

Part number	No of samples required	Tests	Qualifies
M9395/29-ABAAC	2 each resistive	Complete per qualification inspection of MIL-S-9395	All
-GBAAC	2 each resistive		
-A6R3S	2 each inductive		
-N6R3S	2 each inductive		

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - AV
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities
Army - AT
Navy - MC, AS, SH
Air Force - 19

Preparing activity
Air Force - 85

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1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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